Fiscal Year:	FY 2013	**Task Last Updated:**	FY 09/11/2013
PI Name:	Hammond, Timothy M.D.		
Project Title:	Genotypic and Phenotypic Changes in Yeast Related to Selective Growth Pressures Unique To Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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Comments:	Alternate email: Timothy.Hammond@va.gov NOTE: PI previously at Tulane University until 2007 (Ed., 09/2010) Joint appointments Professor, Department of Internal Medicine, Nephrology Division, Duke University School of Medicine, and Staff Physician, Durham VA Health Care System.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	08/01/2010	**End Date:**	07/31/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-135 ; ISS NOTE: Extended to 7/31/2013 per A. Chu/ARC and NSSC information (Ed., 6/03/2012)		
Key Personnel Changes/Previous PI:	June 2012: Add Dr. Ricardo Saban, a subject matter expert in gene array data analysis, as a consultant. Add Dr. Holly Birdsall as a data warden (co-investigator) so all analysis can be performed blinded.		
COI Name (Institution):	Birdsall, Holly (Department of Veterans Affairs)		
Grant/Contract No.:	NNX10AP01G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This application aims to investigate how cells adapt to the unique aspects of the space environment, using the model eukaryotic organism, Saccharomyces cerevisiae. Direct quantitative assessment of selective pressures on cell populations through generations has become possible due to the recent availability of the yeast deletion series. This powerful reagent is a mixture of molecularly engineered isogenic yeast strains that differ only in a single gene locus. Each gene has been replaced with an identifying "bar code" and the mixture contains a deletion strain for every gene in the yeast genome. The yeast deletion series is grown under a selective pressure, DNA extracted, and the barcodes amplified by PCR. The resulting product is annealed to a gene microarray chip comprising spots for the complementary sequence of each barcode. In this manner the selective growth of every strain in the yeast deletion series can be assayed in a single tube, enabling a genomic approach to phenotypic analyses. These studies have a strong link and a clear path to flight research on the International Space Station because of the properties of the yeast deletion series: it can be flown in a quiescent state, and activated by simple fluid addition and completed by fixative addition. Importantly, the post-flight analysis is of DNA, which is orders of magnitude easier to fix and preserve than RNA. Given these characteristics, we propose a flight experiment, with a high probability of obtaining genome wide data. This flight experiment directly addresses two broad areas Fundamental Space Biology is attempting to expand: 1) Life at molecular and cellular levels, & 2) Life across generations To mimic some of the elements of microgravity on the ground, in a proof-of-concept experiment we propose to utilize the rotating wall vessel, a form of suspension culture which mimics some elements associated with microgravity, namely random orientation to the gravitational vector, and falling at terminal velocity. Sets of the rotating wall vessel grown under conventional means will control for rotational effects, and spotting the yeast series on agar plates controls for the effects of suspension culture. Our hypothesis is that yeast cells will undergo a selective response to growth conditions that reflect the space environment and that this phenotypic response will be controlled by changes in gene expression of specific target genes. Based on our preliminary data our analysis will examine both specific pathways such as ones utilizing the Stress Responsive Element (STRE), as well as ones that test for novel pathways. The steps we will take to examine this hypothesis are as follows below: 1. Identify those yeast genes regulating growth under the selective pressure of the rotating wall vessel, using the yeast deletions strains 2. Confirm the ability of the specific regulatory motifs to directly activate transcription of a set of reporter genes 3. Identify those trans-acting factors that interact with these regulatory motifs 4. Confirm and extend the results obtained in our ground based studies on the selective growth pressure in a true microgravity environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space based growth of microorganisms appears to induce a unique set of virulence factors, which may identify gene pathways which can be deleted to generate weakened bacterial forms suitable for vaccine development.
Task Progress:	Genes determining survival advantage in S. cerevisiae changes during static, RWV, and flight plate culture - Deletion series and scanning electron microscopy studies. On the last space shuttle flight, known as STS-135, our labs flew the yeast deletion series grown in suspension culture. Scanning electron micrographs show the phenotype of the yeast grown in true microgravity as compared to ground-based microgravity emulation. Analysis of the gene deletions that mediated growth advantage or disadvantage during rotation or flight allows construction of pathway diagrams to summarize the data. Scanning electron microscopy (10,000X) demonstrated differences in shape, surface texture and budding pattern when grown in YPD in Opticells as suspension cultures. Static ground controls had egg to barrel shaped cells, with single pole budding, and an undulated surface. Rotating wall vessel grown yeast were round to barrel shaped, with multiple, bi-polar budding, and greatly undulated surface. Space flown cultured yeast were round to egg shaped, with multiple budding > ground <RWV; and large smooth patches associated with active budding. The space flight cultured yeast had medium surface undulation. Identification of genetic - environmental interactions in the flown and grown samples versus flown-not grown controls. In this analysis the normalized sizes of each of 4 replicates for each single mutant deletion strain was compared to the flown not-grown samples that were grown upon their return. Pathway analysis. The 50 strains with the greatest space-related fitness defects were imported into the meta-analytical tool, GeneMANIA (http://www.genemania.org). GeneMANIA comprises a series of algorithms that searches large, publicly available biological datasets to find related genes. These include protein-protein, protein-DNA and genetic interactions, pathways, reactions, gene and protein expression data, protein domains and phenotypic screening profiles. These relationships can be of several types: Co-expression: Two genes are linked if their expression levels are similar across conditions in a gene expression study. Physical Interaction: Two gene products are linked if they were found to interact in a protein-protein interaction study. Genetic interaction: Two genes are functionally associated if the effects of perturbing one gene were found to be modified by perturbations to a second gene. Co-localization: Genes expressed in the same tissue, or proteins found in the same location. Two genes are linked if they are both expressed in the same tissue or if their gene products are both identified in the same cellular location. We then prepared a visual representation of the output from this analysis where each gene is a node in the network, and the links between each node represents one type of interaction between these 50 genes. This preliminary analysis reveals several clusters of functionally interconnected genes, including clusters of genes involved in DNA repair and metabolism, transcription, chromatin remodeling, mitochondrial function, polarity, ribosome assembly, and stress.

	In conclusion: yeast deletion series are a powerful tool to study pathways mediating evolution fitness for life. These studies have allowed us to define the genes which convey survival advantage and disadvantage during spaceflight, the pathways which the genes populate, and the control mechanisms which determine the changes.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
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